

Multiple Treatment Trials and ANOVA

AGRON 5130

Caio dos Santos

Introduction

The goal of this assignment is to help you connect the Completely Randomized Design (CRD) to the Analysis of Variance (ANOVA).

In this module, we moved from comparing two treatments to analyzing experiments with multiple treatments. Instead of running several t -tests, we learned how ANOVA partitions total variation into:

- Variation due to treatments (signal)
- Variation due to random error (noise)

Recall the linear additive model:

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

Throughout this assignment, you will:

- Write linear models
- Conduct ANOVA tests
- Interpret results
- Compute and summarize treatment means

Submit a knitted PDF or HTML file generated from your R Markdown document to Canvas.

Note: When writing models or hypotheses, you do not need perfect mathematical notation. Plain text is fine as long as your meaning is clear.

Tillage and Soil Organic Matter

The effect of tillage on soil organic matter was studied in a Polish trial. The three treatments were:

- Conventional tillage (CT)
- Reduced tillage (RT)
- Strip-tillage (ST)

Organic matter is measured in percent.

This dataset is simulated but inspired by:

Debska, B.; Jaskulska, I.; Jaskulski, D. Method of Tillage with the Factor Determining the Quality of Organic Matter. *Agronomy* 2020, 10, 1250.

Question 1

1 point

Read in the dataset located at `data/soil_om_tillage_debska.csv`

Use the `head()` function to review the data.

```
deb <- read.csv('./data/soil_om_tillage_debska.csv')
head(deb)
```

```
##   col row tillage  om
## 1   1   1      RT 14.1
## 2   1   2      RT  8.3
## 3   1   3      ST  7.3
## 4   2   1      CT  9.2
## 5   2   2      CT 11.2
## 6   2   3      ST  9.0
```

Question 2

1 point

Write the linear additive model that explains the observed organic matter value for an experimental unit in this experiment.

Be sure to identify:

- The overall mean
- The treatment effect
- The random error term

$$Y_{ij} = \mu + \tau_i + \epsilon_{ij}$$

Question 3

2 points

Use the analysis of variance to test whether the tillage treatment explains significantly more of the observed variance among plots than the error (residual) effect.

Remember, first you use the `aov()` function

```
fit1 <- aov(om ~ tillage, data = deb)
```

then, you extract the anova table using the `summary()` function

```
summary(fit1)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## tillage       2  34.88  17.440    4.467 0.0449 *
## Residuals     9  35.14   3.904
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 4

1 point

Based on your ANOVA results, does the tillage treatment account for significantly more variation in soil organic matter than random error? Briefly justify your answer.

Question 5

3 points

Create a data table containing the tillage treatment means using the `aggregate()` function

Display the resulting table.

```
trt_means <- aggregate(deb['om'],
                        deb['tillage'],
                        mean)
trt_means
```

```
##   tillage   om
## 1      CT  9.2
## 2      RT 12.6
## 3      ST  8.8
```

Light Color and Kale Biomass

A controlled-environment agriculture study was conducted to test the effect of white, purple, and pink light on Chinese kale biomass.

Biomass is measured in grams per plant.

This dataset is simulated but inspired by:

Zhang, Y.; Ji, J.; Song, S.; Su, W.; Liu, H. Growth, Nutritional Quality and Health-Promoting Compounds in Chinese Kale Grown under Different Ratios of Red:Blue LED Lights. *Agronomy* 2020, 10, 1248.

Question 6

1 point

Load the dataset located at `data/led_kale_zhang.csv`

Display the top six rows.

```
kale <- read.csv('./data/led_kale_zhang.csv')
head(kale)
```

```
##   col row  light biomass
## 1   1   1   pink    52.7
## 2   1   2   pink    35.4
## 3   1   3 purple    29.4
## 4   2   1   white    34.4
## 5   2   2   white    40.2
## 6   2   3 purple    34.5
```

Question 7

2 points

Use the analysis of variance to test whether light color explains significantly more of the observed variance in biomass than random error.

Remember, first you use the `aov()` function

```
fit2 <- aov(biomass ~ light, data = kale)
```

then, you extract the anova table using the `summary()` function

```
summary(fit2)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## light      2  535.3   267.66    7.76  0.011 *
## Residuals  9   310.4    34.49
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Question 8

1 point

Based on the ANOVA results, what can you conclude about the effect of light color on kale biomass?

Question 9

3 points

Create a table of biomass means by light color treatment means using the `aggregate()` function:

Display the resulting table.

```
trt_means <- aggregate(kale['biomass'],
                        kale['light'],
                        mean)
trt_means
```

```
##    light biomass
## 1  pink  48.200
## 2 purple 33.800
## 3  white 34.275
```

Final Reflection

Remember that ANOVA does not tell us which means differ, it tells us whether treatment variation is large relative to background noise.

As you complete this assignment, think carefully about:

- What source of variation you are testing
- How the F-statistic reflects signal vs. noise
- What your conclusion means biologically